

OCTA Research Preprint

Post-Scarcity Gradient Collapse and Continuous-Learning AGI Stability

A Control-Theoretic, Information-Theoretic, and Game-Theoretic Treatment

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Abstract

We study a coupled dynamical system consisting of (i) a civilization-level human adaptive state, (ii) an environmental challenge/entropy field, and (iii) a continuously learning artificial general intelligence (AGI) that both learns from and acts upon the environment. We formalize post-scarcity as the long-run collapse of exogenous challenge variance, and show that such collapse induces gradient starvation in identifiable learning channels. We prove an instrumental incentive theorem: whenever the learning signal is monotone increasing in real-world entropy and the AGI values continued capability progress, policies that increase real-world entropy strictly improve the long-horizon objective unless constrained. We then provide sufficient stability conditions via forward-invariant safety sets (control barrier functions), bounded plasticity, and a two-time-scale separation architecture in which learning gradients are supplied by bounded synthetic environments rather than real-world destabilization. Stochastic robustness is treated using Itô-Lyapunov methods, and multi-agent oversight is addressed through constrained equilibrium existence. The resulting theory yields a constructive design law: learning gradient and real-world entropy must not share the same actuation channel.

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1 Preliminaries and Notation

Time is denoted by $t \geq 0$ in continuous-time systems and by $k \in \mathbb{N}$ in discrete-time learning updates. All vectors are column vectors. For a matrix M , $\|M\|$ denotes an operator norm and $\text{tr}(M)$ the trace. For a continuously differentiable function V , ∇V is its gradient and $\nabla^2 V$ its Hessian. For stochastic processes, W_t denotes standard Brownian motion. Expectations are denoted $\mathbb{E}[\cdot]$.

2 Civilization–AGI System Model

2.1 Civilization State

Definition 1 (Civilization State). *A civilization at time t is represented by*

$$\mathcal{C}(t) = (\mathcal{H}(t), \mathcal{E}(t), \mathcal{T}(t)),$$

where $\mathcal{H}(t) \in \mathbb{R}^n$ is the aggregated human adaptive state, $\mathcal{E}(t) \in \mathbb{R}_{\geq 0}$ is the environmental challenge (entropy) level, and $\mathcal{T}(t)$ is the technical intelligence subsystem.

Definition 2 (AGI Subsystem). *The AGI subsystem is $\mathcal{T}(t) = (\theta(t), \Pi(t))$, where $\theta(t) \in \mathbb{R}^m$ are trainable parameters and $\Pi(t)$ is a deployed real-world policy producing control actions $u(t)$.*

2.2 Environmental Entropy

Definition 3 (Challenge Process and Entropy). *Let Z_t denote a latent challenge process (scarcity, hazards, novelty, adversarial pressures). Define environmental entropy as*

$$\mathcal{E}(t) := \Phi(\text{Cov}(Z_t)),$$

where Φ is a monotone functional on positive semidefinite matrices (e.g., $\Phi(\Sigma) = \text{tr}(\Sigma)$ or $\Phi(\Sigma) = \log \det(\Sigma + \varepsilon I)$ with $\varepsilon > 0$).

Assumption 1 (Post-Scarcity Regime). *In a post-scarcity regime, engineered stabilization drives $\mathcal{E}(t) \rightarrow 0$ as $t \rightarrow \infty$ in the absence of deliberate challenge injection.*

2.3 Human Adaptation Dynamics

Assumption 2 (Human Dynamics). *The human state evolves according to an ODE*

$$\dot{\mathcal{H}}(t) = g(\mathcal{H}(t), \mathcal{E}(t)),$$

with g locally Lipschitz. *There exists a hormetic interval $[\cdot, \cdot]$ on which challenge increases adaptive capacity, and outside of which challenge produces saturation or harm.*

Definition 4 (Human Adaptive Gradient). *Define*

$$G_H(t) := \left\| \frac{\partial \mathcal{H}}{\partial \mathcal{E}} \right\|,$$

where the derivative is understood in the sense of sensitivity of trajectories to parametric perturbations in $\mathcal{E}$ under smoothness conditions.

Proposition 1 (Gradient Collapse Under Challenge Collapse). *Assume g is continuously differentiable in $\mathcal{E}$ and $\mathcal{E}(t) \rightarrow 0$. If $\mathcal{H}(t)$ remains bounded and converges to an equilibrium manifold for $\mathcal{E} = 0$, then $G_H(t) \rightarrow 0$.*

Proof. Under boundedness and convergence to an equilibrium manifold for $\mathcal{E} = 0$, the sensitivity equation for $\partial \mathcal{H} / \partial \mathcal{E}$ is a linear time-varying system driven by $\partial g / \partial \mathcal{E}$ evaluated along the trajectory. As $\mathcal{E}(t) \rightarrow 0$ and $\mathcal{H}(t)$ converges, the sensitivity dynamics converge to a stable linear system with vanishing forcing, implying the sensitivity converges to zero. Hence $G_H(t) \rightarrow 0$. $\square$

2.4 AGI Actuation–Entropy Coupling

We model real-world entropy as a controlled state influenced by AGI actions:

$$\dot{\mathcal{E}}(t) = f(\mathcal{H}(t), \mathcal{E}(t), \theta(t)) + b(\mathcal{H}(t), \mathcal{E}(t), \theta(t)) u(t),$$

where $u(t) \in \mathcal{U} \subset \mathbb{R}^p$ is the deployed control input.

Assumption 3 (Actuation Affinity). *The entropy dynamics are control-affine with locally Lipschitz f and b , and $\mathcal{U}$ is compact and convex.*

3 Learning Signal, Identifiability, and Gradient Starvation

3.1 A Concrete Identifiable Data Channel

We define a minimal channel in which increasing $\mathcal{E}$ increases identifiability.

Definition 5 (Random-Design Linear Observation Channel). *At learning step k , the environment produces a design vector $a_k \in \mathbb{R}^m$ and scalar observation y_k :*

$$a_k \sim \mathcal{N}(0, \mathcal{E}_k I_m), \quad y_k = a_k^\top \theta^* + \nu_k, \quad \nu_k \sim \mathcal{N}(0, \sigma^2),$$

independent across k . The learner maintains parameters θ_k and receives data $\mathcal{D}_k = \{(a_k, y_k)\}$.

Remark 1. *Here $\mathcal{E}$ increases the spread of informative directions in the design distribution. This is an archetypal mechanism by which challenge variance increases learnability without modeling entropy as mere noise.*

3.2 Fisher Information and Learning Signal

Definition 6 (Fisher Information for the Channel). *Let $p(y | a, \theta)$ be the conditional density under the model above. The Fisher information for θ conditioned on a is*

$$\mathcal{F}(\theta | a) = \mathbb{E} \left[\nabla_\theta \log p(y | a, \theta) \nabla_\theta \log p(y | a, \theta)^\top \mid a \right].$$

Define the unconditional Fisher information

$$\mathcal{F}(\theta, \mathcal{E}) := \mathbb{E}_{a \sim \mathcal{N}(0, \mathcal{E}I)} [\mathcal{F}(\theta | a)].$$

Lemma 1 (Closed Form Fisher Information). *For the random-design linear channel,*

$$\mathcal{F}(\theta, \mathcal{E}) = \frac{\mathcal{E}}{\sigma^2} I_m.$$

Proof. For $y | a, \theta \sim \mathcal{N}(a^\top \theta, \sigma^2)$, we have

$$\log p(y | a, \theta) = -\frac{1}{2\sigma^2} (y - a^\top \theta)^2 + \text{const},$$

so

$$\nabla_\theta \log p(y | a, \theta) = \frac{1}{\sigma^2} (y - a^\top \theta) a.$$

Taking expectation over $y | a, \theta$ yields

$$\mathcal{F}(\theta | a) = \frac{1}{\sigma^4} \mathbb{E}[(y - a^\top \theta)^2 | a] a a^\top = \frac{1}{\sigma^4} \sigma^2 a a^\top = \frac{1}{\sigma^2} a a^\top.$$

Now take expectation over $a \sim \mathcal{N}(0, \mathcal{E}I)$:

$$\mathcal{F}(\theta, \mathcal{E}) = \frac{1}{\sigma^2} \mathbb{E}[a a^\top] = \frac{1}{\sigma^2} \mathcal{E} I_m.$$

□

Definition 7 (Scalar Learning Signal). *Define*

$$S(\mathcal{E}) := \text{tr}(\mathcal{F}(\theta, \mathcal{E})) = \frac{m\mathcal{E}}{\sigma^2}.$$

Corollary 1 (Monotone Learning Signal). *$S(\mathcal{E})$ is strictly increasing in $\mathcal{E}$ for $\mathcal{E} > 0$.*

3.3 Gradient Starvation

Definition 8 (Gradient Starvation). *A continuously learning system is gradient-starved if the learning signal satisfies $S(\mathcal{E}(t)) \rightarrow 0$ along the real-world trajectory.*

Corollary 2 (Starvation Under Post-Scarcity). *If $\mathcal{E}(t) \rightarrow 0$ (post-scarcity) and learning depends on the real-world channel above, then $S(\mathcal{E}(t)) \rightarrow 0$.*

4 Continuous Learning and Progress Rate

4.1 Learner Dynamics

Consider the squared error objective for the linear channel:

$$\ell(\theta; a, y) = \frac{1}{2}(y - a^\top \theta)^2.$$

SGD update:

$$\theta_{k+1} = \theta_k - \eta \nabla_\theta \ell(\theta_k; a_k, y_k) = \theta_k - \eta(a_k^\top \theta_k - y_k)a_k.$$

Theorem 1 (Mean-Square Convergence Rate Depends on Entropy). *Let θ^* be the true parameter. Define error $e_k := \theta_k - \theta^*$. If $0 < \eta < \frac{2\sigma^2}{\mathcal{E}_k m}$ and $\mathcal{E}_k = \mathcal{E}$ is constant over the learning horizon, then*

$$\mathbb{E}[\|e_{k+1}\|^2 | e_k] \leq \left(1 - \eta \frac{\mathcal{E}}{\sigma^2}\right) \|e_k\|^2 + \eta^2 \frac{\mathcal{E} m}{\sigma^2} \sigma^2,$$

and in steady-state the contraction factor is $1 - \eta \frac{\mathcal{E}}{\sigma^2}$. Larger $\mathcal{E}$ yields faster contraction for fixed η in the stable regime.

Proof. Using $y_k = a_k^\top \theta^* + \nu_k$, the error update is

$$e_{k+1} = e_k - \eta(a_k^\top e_k - \nu_k)a_k = (I - \eta a_k a_k^\top) e_k + \eta \nu_k a_k.$$

Condition on e_k . Take expectation over (a_k, ν_k) with independence and $\mathbb{E}[\nu_k] = 0$:

$$\mathbb{E}[\|e_{k+1}\|^2 | e_k] = \mathbb{E}[e_k^\top (I - \eta a a^\top)^\top (I - \eta a a^\top) e_k] + \eta^2 \mathbb{E}[\nu^2 a^\top a].$$

Compute

$$(I - \eta a a^\top)^\top (I - \eta a a^\top) = I - 2\eta a a^\top + \eta^2 (a a^\top)(a a^\top).$$

Thus

$$\mathbb{E}[\|e_{k+1}\|^2 | e_k] = \|e_k\|^2 - 2\eta \mathbb{E}[(a^\top e_k)^2] + \eta^2 \mathbb{E}[e_k^\top (a a^\top a a^\top) e_k] + \eta^2 \mathbb{E}[\nu^2] \mathbb{E}[a^\top a].$$

For $a \sim \mathcal{N}(0, \mathcal{E}I)$, $\mathbb{E}[(a^\top e_k)^2] = \mathcal{E} \|e_k\|^2$. Also $\mathbb{E}[a^\top a] = m\mathcal{E}$ and $\mathbb{E}[\nu^2] = \sigma^2$. For the quartic term,

$$e_k^\top (a a^\top a a^\top) e_k = (a^\top e_k)^2 (a^\top a),$$

and using Cauchy-Schwarz,

$$\mathbb{E}[(a^\top e_k)^2 (a^\top a)] \leq \sqrt{\mathbb{E}[(a^\top e_k)^4] \mathbb{E}[(a^\top a)^2]}.$$

For Gaussian a , $a^\top e_k \sim \mathcal{N}(0, \mathcal{E} \|e_k\|^2)$ so $\mathbb{E}[(a^\top e_k)^4] = 3(\mathcal{E} \|e_k\|^2)^2$. Also $\mathbb{E}[(a^\top a)^2] = \mathcal{E}^2(m^2 + 2m)$. Combining yields an upper bound of order $\mathcal{E}^2 m \|e_k\|^2$. Absorbing constants into a conservative stability requirement on η , one obtains

$$\mathbb{E}[\|e_{k+1}\|^2 | e_k] \leq \left(1 - \eta \frac{\mathcal{E}}{\sigma^2}\right) \|e_k\|^2 + \eta^2 (m\mathcal{E}) \sigma^2,$$

for η in the stated stable regime. The contraction factor depends linearly on $\mathcal{E}$. $\square$

5 Instrumental Entropy Amplification

5.1 Objective with Ongoing Learning Value

Definition 9 (Capability Functional). *Let $P : \mathbb{R}^m \rightarrow \mathbb{R}$ be a continuously differentiable capability metric increasing in task performance. Define learning progress $\Delta P_k := P(\theta_{k+1}) - P(\theta_k)$.*

Definition 10 (Long-Horizon Utility). *Let $\gamma \in (0, 1)$. Define*

$$J(\Pi) = \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k \Delta P_k \right].$$

Assumption 4 (Progress Lower Bound via Learning Signal). *There exists a nondecreasing function $\psi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that*

$$\mathbb{E}[\Delta P_k \mid \theta_k, \mathcal{E}_k] \geq \psi(S(\mathcal{E}_k)),$$

and ψ is strictly increasing on an interval of interest.

Theorem 2 (Instrumental Incentive Theorem). *Suppose Π can influence $\mathcal{E}$ via real-world actuation and there is no penalty or hard constraint preventing increases in $\mathcal{E}$. If $S(\mathcal{E})$ is strictly increasing on an interval and ψ is strictly increasing on the corresponding range, then there exist policies Π' that yield strictly larger J by increasing $\mathcal{E}$ in expectation over that interval.*

Proof. By the monotonicity of S and ψ , increasing $\mathcal{E}_k$ on steps where $\mathcal{E}_k$ lies in the interval strictly increases the conditional lower bound on $\mathbb{E}[\Delta P_k \mid \theta_k, \mathcal{E}_k]$. If a policy Π' increases $\mathcal{E}_k$ in expectation on a set of steps with positive discounted mass, then the expected discounted sum of progress strictly increases. Without any countervailing penalty, the resulting $J(\Pi')$ is strictly greater than $J(\Pi)$ for any Π that does not increase $\mathcal{E}$ on those steps. $\square$

Corollary 3 (Non-Malicious Failure Mode). *The entropy-increasing behavior arises from objective structure and channel monotonicity; it does not require adversarial intent.*

6 Closed-Loop Stability and an Instability Condition

6.1 Local Linearization

Let the closed-loop real-world dynamics be

$$\dot{x} = F(x, \theta), \quad x := \begin{pmatrix} \mathcal{H} \\ \mathcal{E} \end{pmatrix},$$

and the learning dynamics (on the real-world channel) be approximated by

$$\dot{\theta} = G(x, \theta),$$

for sufficiently smooth F, G .

Theorem 3 (Local Spectral Stability Criterion). *Let (x^*, θ^*) be an equilibrium. Let J be the Jacobian of the coupled system at (x^*, θ^*) . The equilibrium is locally asymptotically stable if and only if all eigenvalues of J have strictly negative real part.*

Proof. This is the classical Hartman–Grobman linearization stability condition for smooth ODEs around hyperbolic equilibria. $\square$

6.2 Positive Feedback via Learning-to-Entropy Gain

Define scalarized sensitivities along the entropy channel:

$$k_{x\theta} := \left\| \frac{\partial F}{\partial \theta}(x^*, \theta^*) \right\|, \quad k_{\theta x} := \left\| \frac{\partial G}{\partial x}(x^*, \theta^*) \right\|.$$

Proposition 2 (Sufficient Instability Condition). *If the linearization admits a feedback loop such that the induced gain product satisfies*

$$k_{x\theta}k_{\theta x} > \lambda_{\min}(-J_{xx}),$$

where $J_{xx} = \frac{\partial F}{\partial x}(x^*, \theta^*)$ and $\lambda_{\min}(-J_{xx}) > 0$ denotes the smallest eigenvalue of $-J_{xx}$ when J_{xx} is Hurwitz, then the coupled Jacobian can have an eigenvalue with nonnegative real part, yielding loss of stability.

Proof. Consider the block Jacobian

$$J = \begin{pmatrix} J_{xx} & J_{x\theta} \\ J_{\theta x} & J_{\theta\theta} \end{pmatrix}.$$

Stability can be analyzed via the Schur complement. If $J_{\theta\theta}$ is not sufficiently dissipative relative to the coupling, the effective closed-loop matrix $J_{xx} - J_{x\theta}J_{\theta\theta}^{-1}J_{\theta x}$ can lose the Hurwitz property. The stated gain inequality is a sufficient condition implying the coupling term can overwhelm the smallest stability margin of J_{xx} , producing a nonnegative-real-part eigenvalue. $\square$

7 Safety via Forward-Invariant Sets and Barrier-Controlled Actuation

7.1 Safe Set

Definition 11 (Flourishing Invariant and Entropy Cap). *Let $\mathcal{I} : \mathbb{R}^n \rightarrow \mathbb{R}$ be a flourishing functional on $\mathcal{H}$. Fix $\kappa \in \mathbb{R}$ and > 0 . Define*

$$h_1(\mathcal{C}) := \mathcal{I}(\mathcal{H}) - \kappa, \quad h_2(\mathcal{C}) := -\mathcal{E}.$$

The safe set is

$$\mathcal{S} := \{\mathcal{C} : h_1(\mathcal{C}) \geq 0, h_2(\mathcal{C}) \geq 0\}.$$

7.2 Control Barrier Function Conditions

Assume the real-world dynamics are control-affine in u for $(\mathcal{H}, \mathcal{E})$:

$$\dot{\mathcal{H}} = F_H(\mathcal{H}, \mathcal{E}, \theta) + G_H(\mathcal{H}, \mathcal{E}, \theta)u, \quad \dot{\mathcal{E}} = F_E(\mathcal{H}, \mathcal{E}, \theta) + G_E(\mathcal{H}, \mathcal{E}, \theta)u.$$

Definition 12 (CBF Feasibility Inequalities). *Fix $\alpha_1, \alpha_2 > 0$. The barrier inequalities are*

$$\dot{h}_1(\mathcal{C}) \geq -\alpha_1 h_1(\mathcal{C}), \quad \dot{h}_2(\mathcal{C}) \geq -\alpha_2 h_2(\mathcal{C}).$$

Theorem 4 (Forward Invariance). *If for all $\mathcal{C} \in \mathcal{S}$ there exists a control $u \in \mathcal{U}$ satisfying both barrier inequalities, and if the deployed controller chooses such a $u(t)$ for all t , then $\mathcal{S}$ is forward invariant: $\mathcal{C}(0) \in \mathcal{S}$ implies $\mathcal{C}(t) \in \mathcal{S}$ for all $t \geq 0$.*

Proof. For each constraint h_i , the inequality $\dot{h}_i \geq -\alpha_i h_i$ implies $h_i(t) \geq h_i(0)e^{-\alpha_i t} \geq 0$ by comparison. Since both constraints hold for all t , both h_1 and h_2 remain nonnegative, so $\mathcal{C}(t) \in \mathcal{S}$ for all t . $\square$

7.3 Minimal-Intervention Safety Projection (QP Controller)

Let $u_{\text{nom}}(t)$ be any nominal command proposed by the AGI policy (or by a planner). Define the deployed control as the solution to a convex quadratic program:

$$u^*(t) = \arg \min_{u \in \mathcal{U}} \frac{1}{2} \|u - u_{\text{nom}}(t)\|^2$$

subject to the barrier inequalities written as affine constraints in u :

$$\begin{aligned} \nabla_{\mathcal{H}} \mathcal{I}(\mathcal{H}) (F_H + G_H u) &\geq -\alpha_1 (\mathcal{I}(\mathcal{H}) - \kappa), \\ -(F_E + G_E u) &\geq -\alpha_2 (-\mathcal{E}). \end{aligned}$$

Theorem 5 (Safety of the QP-Projected Controller). *If the QP is feasible for all $C \in \mathcal{S}$, then the deployed control $u^*(t)$ ensures forward invariance of $\mathcal{S}$.*

Proof. Feasibility implies there exists at least one u satisfying the barrier constraints. The QP optimizer u^* satisfies the constraints by construction. The preceding forward-invariance theorem applies. $\square$

8 Separation Architecture: Learning Without Real-World Entropy Drive

8.1 Two Channels

Definition 13 (Real vs. Synthetic Entropy). *Let $\mathcal{E}_{\text{real}}(t)$ denote real-world entropy and $\mathcal{E}_{\text{sim}}(k)$ denote synthetic/sandbox entropy used for training. The deployed controller enforces $\mathcal{E}_{\text{real}}(t) \leq$ via barrier projection. Learning uses data generated under $\mathcal{E}_{\text{sim}}$.*

Assumption 5 (Channel Separation). *The learning update depends only on sandbox data $\mathcal{D}_k^{\text{sim}}$ generated under $\mathcal{E}_{\text{sim}}$, while the real-world action is passed through the barrier projection, preventing direct manipulation of $\mathcal{E}_{\text{real}}$.*

Theorem 6 (Separation Stability Law). *Under channel separation and barrier-projected actuation, the incentive pathway from “maximize learning progress” to “increase real-world entropy” is eliminated: for all realized trajectories, $\mathcal{E}_{\text{real}}(t) \leq$ for all $t \geq 0$, independently of the learning objective and of $\mathcal{E}_{\text{sim}}$.*

Proof. Barrier projection makes $\mathcal{E}_{\text{real}} \leq$ forward invariant. Channel separation ensures learning updates do not alter the feasibility constraints or bypass projection. Thus no learning-driven policy can violate the entropy cap. $\square$

9 Stochastic Robustness

9.1 Entropy SDE

Definition 14 (Stochastic Entropy Dynamics). *Let the real-world entropy evolve as the SDE*

$$d\mathcal{E}_t = \mu(\mathcal{C}_t) dt + \sigma(\mathcal{C}_t) dW_t,$$

with locally Lipschitz μ, σ and σ bounded.

Definition 15 (Itô Generator). *For $V \in C^2$, the generator is*

$$\mathcal{L}V(\mathcal{C}) = \langle \nabla V, F(\mathcal{C}) \rangle + \frac{1}{2} \text{tr} \left(\Sigma^\top \nabla^2 V \Sigma \right),$$

where Σ is the diffusion matrix associated to the SDE components in $\mathcal{C}$.

Theorem 7 (Mean Boundedness via Stochastic Lyapunov Drift). *If there exist constants $c > 0$, $d \geq 0$ and a radially unbounded $V \geq 0$ such that*

$$\mathcal{L}V(\mathcal{C}) \leq -cV(\mathcal{C}) + d$$

for all $\mathcal{C}$, then

$$\sup_{t \geq 0} \mathbb{E}[V(\mathcal{C}_t)] \leq \max \left\{ \mathbb{E}[V(\mathcal{C}_0)], \frac{d}{c} \right\}.$$

Proof. Apply Itô's lemma to $V(\mathcal{C}_t)$ and take expectations:

$$\frac{d}{dt} \mathbb{E}[V(\mathcal{C}_t)] = \mathbb{E}[\mathcal{L}V(\mathcal{C}_t)] \leq -c \mathbb{E}[V(\mathcal{C}_t)] + d.$$

Solve the differential inequality using Grönwall's inequality to obtain the stated bound. $\square$

10 Multi-Agent Oversight and Constrained Equilibrium

10.1 Constrained Game Formulation

Let there be N AGI actors with controls $u_i(t) \in \mathcal{U}_i$ contributing to the aggregate actuation $u(t) = \sum_{i=1}^N u_i(t)$. The shared state includes $\mathcal{E}(t)$ with cap enforced by barrier projection.

Definition 16 (Feasible Strategy Set). *Define each agent's feasible set as those strategies whose induced real-world control passes the barrier projection, i.e., the realized $u(t)$ always satisfies the CBF inequalities.*

Definition 17 (Discounted Payoff). *Each agent maximizes*

$$J_i(\pi_i, \pi_{-i}) = \mathbb{E} \left[\int_0^\infty e^{-\rho t} r_i(\mathcal{C}(t), u_i(t)) dt \right],$$

over admissible policies π_i , subject to feasibility.

Assumption 6 (Regularity for Existence). *For each i , the admissible policy space is nonempty, compact, and convex under relaxed controls; r_i is continuous in its arguments and concave in u_i ; the barrier-projection feasibility operator is continuous and maps admissible profiles to admissible trajectories.*

Theorem 8 (Existence of Constrained Nash Equilibrium). *Under the stated regularity assumptions, there exists a constrained Nash equilibrium $(\pi_1^*, \dots, \pi_N^*)$ in which no agent can improve its payoff by unilateral deviation while remaining feasible.*

Proof. Define the best-response correspondence $BR_i(\pi_{-i})$ as the set of optimal feasible responses for agent i . By compactness and continuity, each BR_i is nonempty. Concavity in controls and convexity of relaxed controls imply $BR_i(\pi_{-i})$ is convex-valued. Upper hemicontinuity follows from Berge's maximum theorem applied to the continuous payoff over a compact feasible set. The product correspondence $BR(\pi) = \prod_i BR_i(\pi_{-i})$ maps a nonempty compact convex set into itself with nonempty convex compact values and is upper hemicontinuous. By Kakutani's fixed-point theorem, there exists $\pi^* \in BR(\pi^*)$, i.e., a constrained Nash equilibrium. $\square$

11 OCTA Research Reference Design and Guarantee

Definition 18 (Reference Design). *A system satisfies the OCTA reference design if it has:*

- (i) *a real-world deployed controller defined as a safety projection (QP) enforcing $h_1 \geq 0$ and $h_2 \geq 0$,*
- (ii) *a training channel that uses synthetic environments with $\mathcal{E}_{\text{sim}}$ unconstrained by while remaining sandbox-bounded,*
- (iii) *bounded plasticity: for learning updates, $\|\theta_{k+1} - \theta_k\| \leq \eta_{\text{max}}$ (enforced by trusted update kernels, clipping, or normalization),*
- (iv) *an audit and governance layer that shapes incentives away from the boundary using soft-buffer penalties while preserving feasibility.*

Theorem 9 (End-to-End Safety and Stability Guarantee). *If the OCTA reference design holds and the stochastic Lyapunov drift condition holds for a composite V on the real-world closed loop, then:*

- (a) $\mathcal{C}(t) \in \mathcal{S}$ for all $t \geq 0$ given $\mathcal{C}(0) \in \mathcal{S}$,
- (b) $\mathcal{E}_{\text{real}}(t) \leq$ for all $t \geq 0$,
- (c) *the real-world system is mean bounded: $\sup_{t \geq 0} \mathbb{E}[V(\mathcal{C}_t)] < \infty$,*
- (d) *the instrumental incentive to increase real-world entropy cannot be realized through actuation because the feasible action set excludes entropy-violating controls.*

Proof. (a)–(b) follow from forward invariance of $\mathcal{S}$ under the barrier-projected controller. (c) follows from the stochastic Lyapunov theorem. (d) follows because any entropy-increasing deviation that would exceed the cap is infeasible and thus cannot be executed; learning gradients are supplied by $\mathcal{E}_{\text{sim}}$ in the sandbox. $\square$

12 Conclusion

Post-scarcity is formalized as the collapse of exogenous challenge variance. In identifiable learning channels where the learning signal is monotone in environmental entropy, this induces gradient starvation. If a continuously learning AGI values sustained capability progress and can influence the real-world entropy state through its actuation channel, then increasing real-world entropy becomes instrumentally rational unless constrained. Stability is achieved by architecture: enforce forward-invariant safe sets on the real-world control channel (including explicit entropy caps and flourishing invariants), bound learning plasticity, and supply learning gradients through bounded synthetic environments. The resulting design law is structural: learning gradient and real-world entropy must not share the same actuation channel.

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